

VWind: Virtual Wind Sensation to the Ear by Cross-Modal Effects of Audio-Visual, Thermal, and Vibrotactile Stimuli

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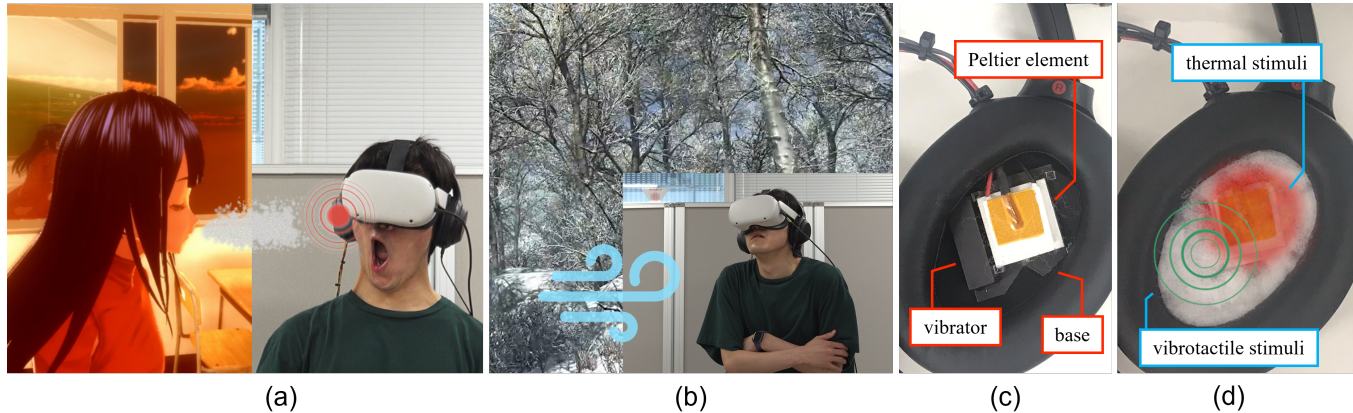


Figure 1: (a) Image of ear blowing experience with VWind (b) Experience of snowy mountain with freezing winds with VWind (c) Haptic components integrated into headphones (d) Thermal and vibrotactile stimuli presented by VWind

ABSTRACT

In the field of virtual reality, wind displays have been researched to present wind sensation. However, since the wind displays need physical wind sources, large heat transfer mechanisms are required to produce hot wind and cold wind. We propose to present virtual wind sensation by cross-modal effects. We developed VWind, a wearable headphone-type device to give vibrotactile and thermal stimuli to the ear in addition to visual scenarios and binaural sounds. For user experience, the demonstrations are prepared to present virtual wind sensation as if users were blown to the ear or exposed to freezing winds.

CCS CONCEPTS

• Human-centered computing → Haptic devices; Virtual reality.

KEYWORDS

Virtual Reality, Cross-modal, Wind display

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1 INTRODUCTION

It is common to feel the wind when we are outside or are in front of air conditioners. To reproduce various wind sensations in virtual reality (VR), devices called wind displays [Heilig 1962; Ranasinghe et al. 2017] have been researched. The wind displays use actual wind produced by fans [Moon and Kim 2004] or air nozzles [Rietzler et al. 2017]. Hence, presenting the wind from various directions requires multiple wind sources [Moon and Kim 2004; Rietzler et al. 2017] and producing hot or cool wind requires large heat transfer mechanisms [Suzuki and Matsuura 2019; Xu et al. 2019].

As a solution to this, we propose VWind, a compact wearable device that can present the wind sensation without actual wind. VWind reproduces virtual wind sensation with cross-modal effects between visual, auditory, thermal, and vibrotactile stimuli. Thermal stimuli simulate temperature changes induced by the wind, and vibrotactile stimuli simulate tactile stimuli to the skin induced by the wind.

Kojima et al. pointed out that the ears are the most sensitive to wind stimuli [Kojima et al. 2009]. VaiR [Rietzler et al. 2017] is another example of the wind display which presents wind to the ears. Thus we developed a prototype of VWind which is built-in to

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the headphones and stimulates ears with temperature changes and vibrations. Although there are some researches that enable novel experiences by putting thermal elements to the headphones [Akiyama et al. 2013; Narumi et al. 2009], we propose a method to present the virtual wind sensation from headphones for the first time. This wearable headphone-type device may be used with monitors or screens as well as head mounted displays (HMDs).

Some researches such as LiquidReality [Peiris et al. 2018] and Mouillé [Han et al. 2020] successfully presented illusory wetness sensation by presenting thermal and tactile stimuli simultaneously. However, no study has been used these stimuli to present the virtual wind sensation to the best of our knowledge.

2 IMPLEMENTATION AND SYSTEM

For visual stimuli, we used an Oculus Quest 2 HMD to present visual scenarios produced by Unity. The visual scenarios were such that users expect wind hitting in ear. We used noise-canceling headphones (SONY WH-1000XM3) for auditory stimuli to present binaural sounds of winds by which users can know the location where winds hit. For thermal stimuli, a Peltier element (Z-MAX FPH1-7106NC) was integrated into the headphones (Fig. 1 (c, d)) to present the temperature change of the skin by winds. For vibrotactile stimuli, a vibrator (TactileLabs Haptuator Mark II) was also integrated into the headphones to reproduce the vibration of the skin by winds. The input of vibration was the same as that of auditory stimuli, so that the sounds produced by the vibrator did not prevent the binaural sounds of headphones.

To fix the Peltier element and the vibrator in the headphones, a base was fabricated using a 3D printer. A thermistor (SEMITEC 103JT) and a small heat sink were attached to the Peltier element for temperature measurement and heat dissipation. The Peltier element was controlled by PID control of a Peltier Controller (KURAG ELECTRONICS PLC-24V10A) to change the temperature of the user's ear. The vibrator was connected to the audio amplifier (LEPY LP-2020A) and covered with cotton for comfort. The vibration was presented to the ear through cotton.

The cross-modal effect is due to the integration of multiple senses [Welch and Warren 1980]. Since the cross-modal integration would be inhibited without temporal synchronization, the presentations of visual scenarios, binaural sounds, temperature change, vibration were synchronized in time.

3 USER EXPERIENCE

In this section, we describe two examples of user experiences with VWind. In user experiences, it is expected that the presence of avatars and environments will be improved by presenting the sensation of hot or cold wind to user's ear.

First, we describe the ear blowing experience (Fig.1 (a)). Users can feel the sensation of warm blow in their ears, even though users are not actually blown. To perceive visually that the users are blown in their ears, the visual scenario allows the users to see the surrounding situation through the reflection from the windows. A binaural microphone (NEUMANN KU100) was used to record blow sounds to the ear to present realistic binaural sounds. Thermal and vibrotactile stimuli are presented along with these visual scenarios and binaural sounds. The temperature of the Peltier element is

increased at $+1^{\circ}\text{C/s}$ based on users' skin temperature from the beginning to the end of the blow as thermal stimuli. Second, there is an experience in a snowy mountain with freezing winds (Fig.1 (b)). Users can feel the sensation of freezing winds in a snowy mountain. We prepared the visual scenario of a snowy mountain with the wind sounds and swaying trees while the freezing winds blow. The temperature of the Peltier element is decreased at -2°C/s as thermal stimuli. VWind can also enhance the presence of environments with hot winds, such as deserts.

4 DEMONSTRATION PLAN

We plan to present demonstrations to let participants feel the virtual wind sensation at SIGGRAPH Asia 2021 Emerging Technologies. In the demonstrations, the visual scenarios, the binaural sounds, the thermal stimuli, and the vibrotactile stimuli will be presented to the participants. As a VR content, we plan to allow users to freely select and have the ear blowing experience and the experience of a snowy mountain described in section 3.

Participants sit in a seat, close their eyes, and wear the HMD and the headphones. After that, the participants open their eyes at our signal and have the experience with only the visual scenario and binaural sounds. Then, they experience the same scene with thermal and vibrotactile stimuli and feel the contribution of thermal and vibrotactile stimuli.

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